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What's new from component manufacturers

Need for speed is ever increasing in the race of technological development. Samsung has escalated the race to faster mobile processors by promising its first ARM-based design to be built on a 45nm assembly process. STMicroelectronics claims to have launched the world's thinnest accelerometer. Toshiba is in the race for more power-efficient and higher-capacity flash memories.

ed with assisted GPS (A-GPS). It provides low-noise RF front and multi-path mitigation, which helps in reducing errors in urban areas. Assisted GPS takes assistance to discover its position from the GPS equipment at the cellphone towers that maintain constant connection with the GPS satellite. Thus the GPS receiver on the iPhone need not be in always-'on' mode.

Accelerometer. Changing the orientation from landscape to portrait and vice versa is possible because of STMicroelectronics' LIS331 DL accelerometer. Use of this chip has given platform to many application developers for innovations in gaming, photography, etc.

Audio. Wolfson WM6180C is the audio codec that replaces the original WM8758. Audio formats supported by iPhone 3G are AAC, Protected AAC, MP3, MP3 VBR, Audible (formats 2, 3 and 4), Apple Lossless, AIFF and WAV.

Wireless capability. The two chips at the back of the main board are Marvell's single-chip 88W8686 offering Wi-Fi and CSR Bluecore6-ROM.

Camera. iPhone 3G comes with the same 2-megapixel camera as in iPhone 2G, while iPhone 3GS comes with a 3MP camera that is capable of VGA video recording.

It's about putting the best together...

Just like an iPhone which is a combination of different technologies and expertises to produce a world-class effort, this article is an effort to give accurate information from different individuals and agencies. Thanks to Zion Technology, Delhi, where we disassembled the iPhone 2G and 3G. Firms like Semiconductors Insights, Techonline and ifixit cracked the mystery of most of the chips used in the iPhone.

Many experts have experimented their way with an iPhone. For instance, a Japanese hobbyist has turned his iPhone 3GS into a humanoid robot by wiring it to a mechanical body, where the iPhone 3GS acts as the controller. Ultimately, it is the combination of human interface, speed and applications that really counts for a stylish product like an iPhone to be different. You too can use your creativity in these areas to come up with a product that would compete with an iPhone. ●

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